Bing-Xue Zhuang
Ph.D. Student
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Education

2024-Present Ph.D. in Atmospheric and Oceanic Sciences

2019-2021 M.S. in Atmospheric Sciences

Department of Atmospheric Sciences, National Central University, Taiwan

Thesis: A Novel Approach to Assimilate Polarimetric Parameters with an LETKF System: Analysis and Forecast in Real Summer Cases

2015-2019 B.S. in Atmospheric Sciences

Department of Atmospheric Sciences, National Central University, Taiwan

Research Interests

1. **Data Assimilation**: Assimilating radar, profiler, and sounding data with EnKF or 3DVAR to improve the accuracy of quantitative precipitation forecasts.
2. **Numerical Weather Prediction**: Estimating model uncertainties and sensitivities with ensemble predictions for observation targeting.
3. **Cloud Microphysics**: Modifying microphysics parameterization schemes to better illustrate microphysical processes in numerical models.
4. **Applications of Dual-Polarimetric Radars**: Using dual-polarimetric parameters to validate the performance of numerical models and to initiate hydrometeors via data assimilation.
5. **Artificial Intelligence**: Training AI models to learn the relationship between environmental conditions and presence of cloud and to enhance the performance of data assimilation.

Professional Experience

2024-Present *Part-Time Student Employee, Environment Canada and Climate Change, QC, Canada*

- ✧ Collaborating with Data Assimilation and Satellite Meteorology Section to develop AI models adjusting environmental conditions with cloud features

2022-2024 *Research Assistant*, National Central University, Taiwan

- ✧ Following the results of my master’s thesis to complete the development of the novel approach to assimilate dual-polarimetric parameters
- ✧ Investigating the benefits of assimilating advanced profiler observations with the observation system simulation experiment (OSSE)
- ✧ Using ensemble simulation to evaluate the potential locations for the deployment of the advanced observation instrument in the future

2019-2021 *Graduate Research Assistant*, National Central University, Taiwan

- ✧ Upgraded the EnKF radar data assimilation system by adding the function of assimilating dual-polarimetric parameters

- ✧ Evaluated the benefits and limitations of assimilating dual-polarimetric parameters with different microphysics parameterization schemes
- ✧ Tested a novel approach to take advantage of the relationship between Z_{DR} and the mean diameter of raindrops in the data assimilation procedure

2018 July *Summer Internship*, Central Weather Bureau, Taiwan

- ✧ Built a global numerical model with the barotropic vorticity equation
- ✧ Investigated the energy dispersion of an ideal typhoon vortex

Honors and Awards

2025 Pilarczyk Fellowship, CAD 5000 (≈ USD 3500)

2021 Taiwan Rotary Academic Scholarship, NTD 135000 (≈ USD 4500).

2019 National Central University × Hiroshima University Poster Symposium, the 3rd prize award

2019 Ren-Jie Liu Scholarship for Undergraduate Students NTD 20000 (≈ USD 650)

Publications

1. **Zhuang, B.-X.**, K.-S. Chung, W.-Y. Chang, C.-C. Tsai, and Y.-C. Yu: An Innovative Approach to ZDR Data Assimilation Using an Ensemble Kalman Filter: A Proof-of-Concept Study. Atmospheric Research, **332**, 108703, doi: https://doi.org/10.1016/j.atmosres.2025.108703.

Conference Presentations

1. Chung, K.-S., **B.-X. Zhuang**, W.-Y. Chang, and C.-C. Tsai: The Mean Diameter Update Approach for Ensemble-Based Dual-Polarimetric Radar Data Assimilation. 12th European Conference on Radar in Meteorology and Hydrology, Rome, Italy, September 9–13.
2. **Zhuang, B.-X.,** K.-S. Chung, W.-Y. Chang, and C.-C. Tsai: The Mean Diameter Update Approach for Dual-Polarimetric Radar Data Assimilation Based on Ensemble Kalman Filter. AOGS Annual Meeting 2022, Pyeongchang, Korea, June 24–28, 2024
3. **Zhuang, B.-X.**, C.-Y. Ke, Y.-C. Liou, K.-S. Chung, and W.-Y. Chang: Assimilate Planetary Boundary Layer Vertical Profilers in Observation System Simulation Experiments: Impacts on the Quantitative Precipitation Forecast. Conference on Weather Analysis and Forecasting 2023, Taipei, Taiwan, September 5–7, 2023.
4. Chung, K.-S., **B.-X. Zhuang**, W.-Y. Chang, and C.-C. Tsai: A Novel Approach to Assimilate Z_{DR} Observations with an Ensemble Data Assimilation System. Invited Talk Presented at Conference on Weather Analysis and Forecasting 2023, Taipei, Taiwan, September 5–7, 2023.
5. **Zhuang, B.-X.**, K.-S. Chung, W.-Y. Chang, and C.-C. Tsai: A Novel Approach to Assimilate Z_{DR} Observations with an Ensemble Kalman Filter Data Assimilation System, 40th AMS Conference on Radar Meteorology, Minneapolis, Minnesota, USA, August 28–September 1, 2023.
6. **Zhuang, B.-X.**, K.-S. Chung, and C.-C Tsai.: Assimilating Dual-polarimetric Radar Observations with EnKF: Analysis and Forecasts in a Real Case, AOGS Annual Meeting 2022, Virtual, August 1–5, 2022.
7. **Zhuang, B.-X.**, K.-S. Chung, and C.-C Tsai.: Impact of Additional Assimilation of Dual-Polarimetric Parameters: Analysis and Forecasts in a Real Case, EGU General Assembly 2022, Vienna, Austria, May 23–27, 2022, EGU22-4756, https://doi.org/10.5194/egusphere-egu22-4756, 2022.
8. Chung, K.-S., **B.-X. Zhuang**, and C.-C. Tsai: Impact of Assimilating Additional Dual-Polarimetric

Parameters with an LETKF Assimilation System in a Real Summer Case, Conference on Weather Analysis and Forecasting 2021, Taipei, Taiwan, October 26–28, 2021.

9. **Zhuang, B.-X.**, K.-S. Chung, and C.-C. Tsai: Impact on Analysis Fields with Additional Assimilation of Polarimetric Parameters: Two Real Cases Study, AOGS Annual Meeting 2021, Virtual, August 2–6, 2021.
10. **Zhuang, B.-X.**, C.-Y. Lu, P.-J. Chen, K.-S. Chung, and C.-H. Chen: Investigating the Performance of Torrential Rain Event with Different Microphysics Schemes: Case Study of SoWMEX IOP#8, NCU × HU Poster Symposium, Taoyuan, Taiwan, June 10, 2019.

Field Campaign Participation

1. Prediction of Rainfall Extremes Campaign in the Pacific (PRECIP), 2022
   - ✧ Monitored the launch of automatic soundings
   - ✧ Operated SPOL radar and analyzed the real-time data to support PIs' decisions on scanning strategies
2. Taiwan-Area Heavy Rain Observation and Prediction Experiment (TAHOPE), 2022
   - ✧ Operated Taiwan Experimental Atmospheric Mobile Radar (TEAM-R)
   - ✧ Designed scanning strategies depending on weather systems and analyzed the real-time data

Skills

1. **OS**: Windows, macOS, Linux
2. **Programming**: Fortran, Python, NCL, Shell Script
3. **Languages**: Chinese (native), English